A Multi-Architecture Neural Operator Digital Twin: A Proof-of-Concept Surrogate Methodology for Cold-Leg LOCA Screening in AP1000 Nuclear Reactors

Punyansh Thakur, Anushka Paul, Ayush Kumar Bhadra, and Vinay Kumar

***Abstract*—Real-time detection of Cold-Leg Loss-of-Coolant Accidents (LOCA) in pressurized water reactors is constrained by the cost of high-fidelity Computational Fluid Dynamics (CFD) simulation: a single ANSYS Fluent scenario requires 1–4 hours, far too slow for operational screening. Neural operators offer a promising alternative, but existing Deep Operator Network (DeepONet) surrogates suffer from spectral bias, lack physical inductive biases, and have not been compared systematically across architectural paradigms. This paper presents a proof-of-concept digital twin framework for Cold-Leg LOCA screening in the Westinghouse AP1000 reactor, developing and evaluating three neural operator architectures under a synthetic, physics-consistent benchmark: (i) an upgraded *DeepONetFourier* incorporating random Fourier feature encoding, per-neuron adaptive GELU activations, a Sobolev gradient-enhanced loss, and a divergence-free physics penalty; (ii) a *Transolver++* transformer-based operator that tokenizes the computational mesh into $M = 64$ learned tokens for geometry-agnostic, attention-based prediction; and (iii) a *Clifford Neural Operator* built on the Cl(3,0) geometric algebra, providing intrinsic rotational equivariance with only ∼1,800 parameters. All three architectures map input parameters (velocity, break size, temperature) to four coupled thermohydraulic fields over a 25,000-node mesh. Under the synthetic benchmark, DeepONetFourier converges to a validation loss of 9.11×10^{-4} with $R^2 > 0.90$ across all fields, though this result benefits from a stronger loss formulation than the other two architectures receive and should be read as a preliminary inductive-bias observation rather than a controlled benchmark result. A downstream gradient boosting classifier operating on NPPAD-mapped system signals reaches ROC-AUC > 0.95 and recall > 0.92; because these signals are substantially informed by the known input break-size parameter, this result is best read as an indicator-translation demonstration pending independent, field-only validation. End-to-end inference completes in under 20 ms—a ∼180,000× speedup over an approximately 1-hour Fluent reference run. These results establish upper-bound methodological feasibility under controlled, physics-consistent synthetic conditions; full-fidelity CFD validation remains necessary before any operational deployment claim can be made. We provide a preliminary cross-architecture comparison on accuracy, parameter efficiency, inference speed, and physics preservation, offering practitioners an initial trade-off analysis for surrogate selection in nuclear safety monitoring.**

***Index Terms*—Digital Twin, DeepONet, Transolver, Clifford Neural Operator, Geometric Algebra, Transformer Neural Operator, LOCA, AP1000, Nuclear Safety, CFD Surrogate, Fourier Feature Encoding**

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I. Introduction

The AP1000 is a Generation III+ pressurized water reactor (PWR) designed by the Westinghouse Electric Company, featuring 1,000 MWe output and passive safety systems. A Cold-Leg Loss-of-Coolant Accident (LOCA)—a rupture in the primary coolant return piping—is a design-basis event that demands rapid detection to prevent core uncovery and fuel damage [10]. Traditional safety monitoring relies on threshold-based single-sensor alarms, which respond slowly to incipient partial breaks and can miss the early stages of a developing event before critical parameters are exceeded [11].

High-fidelity CFD simulations (e.g., ANSYS Fluent) can predict the complete thermohydraulic state but require 1–4 hours per scenario, making real-time deployment infeasible. Neural operators—machine learning models that learn mappings between infinite-dimensional function spaces—have emerged as powerful Partial Differential Equation (PDE) surrogates. Lu et al. [1] introduced Deep Operator Networks (DeepONet), decomposing the solution operator $\mathcal{G} : \mathcal{U} \to \mathcal{V}$ via a branch-trunk factorization. The baseline DeepONet architecture, however, suffers from well-documented limitations: *spectral bias* causing poor resolution of sharp gradients [7], fixed ReLU activations that cannot adapt to spatially varying fields, and no enforcement of physical conservation laws during training. [Input/confirmation required from the students: the speedup figure reported later in the paper (Section VIII, ∼180,000×) is computed against a ∼3,600 s (≈1 hour) reference run, i.e., the low end of the 1–4 hour range just quoted. Please confirm which Fluent runtime was actually measured/assumed for the speedup calculation and reconcile the two figures; if 1 hour was a deliberately conservative (lower-bound) choice, state that explicitly where the speedup is first reported.]

Subsequent work has explored alternative neural operator paradigms: Transolver [5] applies transformer attention to mesh-compressed tokens for geometry-agnostic PDE solving, and Clifford Neural Operators [6] embed fields in geometric algebra to achieve rotational equivariance architecturally. To the best of our knowledge, no prior published work has systematically examined either architecture for nuclear safety surrogate applications or coupled them to an end-to-end

LOCA screening pipeline, even at the proof-of-concept level. [Input/confirmation required from the students: re-verify this novelty claim against current literature immediately before submission — run a targeted search (e.g., Scopus/Web of Science/arXiv) for "Transolver nuclear," "Clifford neural operator reactor," and "neural operator LOCA" published within the last 12 months, and update this sentence accordingly.]

The Fourier Neural Operator (FNO) [4] is a natural candidate for comparison; however, the fixed structured AP1000 cold-leg geometry and the low-dimensional parametric input (v, b, T) motivated branch–trunk operator learning as the primary architectural comparison focus in this initial study. FNO comparison under appropriate structured-grid conditions is reserved for future controlled benchmarking.

The main contributions of this paper are summarized as follows:

1) We design and implement three neural operator architectures—upgraded DeepONetFourier, Transolver++, and Clifford Neural Operator—within a unified proof-of-concept digital twin pipeline for LOCA screening, each addressing a distinct limitation of the baseline DeepONet.
2) We introduce four improvements over baseline DeepONet—random Fourier feature encoding ($\sigma = 10.0$, $m = 256$), per-neuron adaptive GELU activations, Sobolev gradient loss ($\beta = 0.1$), and a divergence-free penalty ($\lambda = 0.01$)—reducing parameter count by 25–40% while improving convergence under the synthetic benchmark.
3) We apply Transolver++ (mesh-token attention with $M = 64$ tokens, $\mathcal{O}(M^2)$ complexity) and Clifford Neural Operator (Cl(3,0) geometric product, $\sim$1,800 parameters) to reactor flow-field surrogate prediction for the first time in the nuclear domain, to the best of our knowledge.
4) We provide a preliminary comparison of all three architectures on accuracy, parameter efficiency, inference speed, and physics preservation under the present synthetic benchmark and training setup, framed as initial inductive-bias observations rather than a controlled benchmark.
5) We couple the surrogate models to a LOCA indicator-translation demonstration achieving ROC-AUC > 0.95 on NPPAD-mapped signals, with end-to-end model inference in < 20 ms.

The remainder of this paper is organized as follows. Section II reviews related work on neural operators and machine learning for nuclear power plant safety. Section III formulates the surrogate-modeling problem and its input-output specification. Section IV describes the end-to-end system architecture and the five-stage digital twin pipeline. Section V details the three neural operator architectures and their respective improvements over baseline DeepONet. Section VI presents the downstream LOCA indicator detection pipeline, including the NPPAD feature mapping and severity model. Section VII describes implementation details, software, hardware, and the training protocol. Section VIII reports experimental results under the synthetic benchmark. Section IX discusses the methodological contributions, architectural trade-offs, and the limitations that define the path to full-fidelity validation. Section X concludes the paper.

II. Related Work

A. Neural Operators and DeepONet

Neural operators learn mappings between infinite-dimensional function spaces [3]. Two architectures dominate the field: the Fourier Neural Operator (FNO) [4], which operates in the spectral domain with linear mode complexity but requires structured grids, and DeepONet [1], [2], which factorizes the solution operator into branch (input functions) and trunk (output domain) networks via the universal approximation theorem for operators and remains mesh-agnostic. Both suffer from spectral bias—preferentially fitting low-frequency components—a limitation addressed by Fourier feature encoding [8] and Sobolev training [9].

B. Transformer and Geometric Neural Operators

Transolver [5] compresses N-node meshes to $M \ll N$ learned tokens via soft-assignment attention, achieving $\mathcal{O}(M^2)$ complexity independent of mesh size; GNOT [17] applies a similar transformer-based approach to operator learning. Clifford Neural Operators [6] take a different route, embedding physical fields in the Clifford algebra Cl(3,0) to endow the network with rotational equivariance—a mathematically guaranteed inductive bias for fluid mechanics rather than a learned one. Neither architecture has previously been applied to nuclear reactor safety surrogate modeling.

C. Machine Learning for Nuclear Power Plant Safety

Prior work has applied Support Vector Machines (SVMs) [12] and deep recurrent networks [13] to LOCA classification using system-level signals. Digital twin frameworks for nuclear power plants (NPPs) [14], [15] have mostly relied on validated thermal-hydraulics codes (RELAP5, TRACE). Our work bridges high-fidelity CFD surrogates with ML classifiers in a unified proof-of-concept pipeline, and, subject to the literature-verification note above, is among the first to preliminarily compare multiple neural operator architectures for this domain under a synthetic benchmark.

D. Reliability-Engineering Perspective on AI-Based Detection

The present study is framed primarily as a surrogate-modeling and architecture-comparison contribution; it does not yet engage substantively with the reliability-engineering literature on detection-threshold selection, probability-of-detection/false-alarm-rate trade-offs, or verification-and-validation guidance for AI/ML components embedded in safety-critical monitoring systems. [Input/confirmation required from the students: add a dedicated review of (i) probability-of-detection / false-alarm-rate literature for fault- and anomaly-detection systems, (ii) regulatory or standards-body guidance on AI/ML verification and validation for nuclear or other safety-critical applications, and (iii) prior digital-twin-for-NPP work using validated system codes (RELAP5/TRACE) with a direct comparison of scope and validation maturity to the present work.]

III. Problem Formulation

Let $\Omega \subset \mathbb{R}^3$ denote the cold-leg geometry (pipe segment with 90° elbow, diameter $D = 0.7\,\text{m}$, total length $10D$). Define the parameter vector $\boldsymbol{u} = [v, b, T]^\top \in \mathbb{R}^3$, where $v \in [4.0, 6.0]$ m/s is the inlet velocity, $b \in [0, 10]\%$ is the pipe break size, and $T \in [290, 320]$°C is the coolant temperature.

The goal is to learn a neural operator $\mathcal{G}_\theta$ that approximates the steady-state CFD solution operator:

$$\mathcal{G}_\theta(\boldsymbol{u})(\boldsymbol{y}) = \hat{\boldsymbol{s}}(\boldsymbol{y}) = [\hat{p},\ |\hat{\boldsymbol{v}}|,\ \hat{k},\ \hat{\tilde{T}}]^\top \in \mathbb{R}^4 \tag{1}$$

for every spatial location $\boldsymbol{y} \in \Omega$, evaluated simultaneously across $N = 25{,}000$ mesh nodes: output $\hat{\boldsymbol{S}} \in \mathbb{R}^{B\times 4\times N}$, where the reported velocity field is the scalar magnitude $|\hat{\boldsymbol{v}}|$. The divergence-free physics penalty introduced in Section V-A (Eq. 15) requires the full vector field $\hat{\boldsymbol{v}} = (\hat{v}_x, \hat{v}_y, \hat{v}_z)$, not its magnitude; accordingly, the branch-trunk network internally predicts the three vector components $(\hat{v}_x, \hat{v}_y, \hat{v}_z)$ as intermediate outputs, from which the reported scalar $|\hat{\boldsymbol{v}}| = \sqrt{\hat{v}_x^2 + \hat{v}_y^2 + \hat{v}_z^2}$ is derived, and the divergence penalty operates on the intermediate vector prediction. [Input/confirmation required from the students: confirm that the branch-trunk implementation does in fact predict $(\hat{v}_x, \hat{v}_y, \hat{v}_z)$ internally before reducing to $|\hat{\boldsymbol{v}}|$, and specify the corresponding output layer width/shape here and in Section V-A. If instead only the scalar magnitude is predicted, the divergence-free penalty as currently formulated is not computable and Eq. 15 must be reformulated or removed.]

A. CFD Simulation Setup

Fluid properties: pressurized water at 720 kg/m^3, 15.5 MPa operating pressure. The ReNormalization Group (RNG) k-ε turbulence model is used. Boundary conditions: velocity-inlet ($v \in [4.0, 6.0]$ m/s), pressure-outlet (15.5 MPa), no-slip adiabatic walls. A physics-consistent mock data generator produces synthetic CFD fields with pressure gradients, boundary-layer profiles, turbulence peaks at the elbow, and temperature distributions. The dataset comprises 2,000 parameter combinations: 20 velocity × 10 break size × 10 temperature levels, split 70/15/15% into training (1,400), validation (300), and test (300) samples.

Accordingly, the present benchmark should be interpreted as a parametric field-regression proxy with operator-learning structure rather than a fully validated PDE-solution benchmark.

IV. System Architecture and Pipeline

The digital twin operates as a five-stage cascade pipeline (Fig. 1), with each stage transforming data from one physical or learned representation into the next, culminating in a LOCA probability score for screening purposes. A master controller orchestrates the cascade and supports runtime selection of any neural operator architecture through a unified configuration interface.

A. Stage 1: Parametric CFD Data Generation

The training data originates from a parametric sweep of the three-dimensional input space (v, b, T). Two data generation pathways are supported:

Pathway A — ANSYS Fluent (Production): A `FluentSimulationGenerator` module creates Fluent journal files for each parameter combination. Each journal sets velocity-inlet boundary conditions, configures break geometry (modeled as a pressure-outlet with break diameter $d_{\text{break}} = (b/100) \times D$), defines fluid properties, runs steady-state Reynolds-Averaged Navier–Stokes (RANS) simulation with the RNG k-ε model for 1,000 iterations, and exports a CSV with columns $(x, y, z, p, |\boldsymbol{v}|, v_x, v_y, v_z, k, T)$ per node. Production runtime: ∼1–4 hours per simulation.

Pathway B — Physics-Consistent Mock Generator (Development): A synthetic data generator produces fields on a shared 25,000-node mesh (cylindrical pipe, diameter $D = 0.7\,\text{m}$, length $10D$) incorporating:

- **Pressure**: Linear axial gradient ($\Delta p = 50{,}000 \cdot v/5$ Pa) + localized Gaussian break drop ($200{,}000 \cdot b_{\text{norm}}$ Pa, $\sigma = 0.8\,\text{m}$) + radial centrifugal effect ($5{,}000(1 - r^2)$ Pa)
- **Velocity**: Parabolic profile $v(r) = v_{\text{in}}(1 - r^\alpha)$, with $\alpha = 2 - 0.8 b_{\text{norm}} e^{-\frac{(x - x_{\text{break}})^2}{2\sigma^2}}$ (flattens near break) and 30% reduction at full break
- **Turbulence**: Base $k = 0.01v^2$, wall-layer increase (factor $1 + 1.5r$), break-induced spike ($5b_{\text{norm}}$, localized at break)
- **Temperature**: Kelvin base + radial gradient ($3(1 - r^2)$ K) + downstream break hot-spot ($15 b_{\text{norm}}$ K) + axial cooling (2 K over pipe length)

where $b_{\text{norm}} = b/10 \in [0, 1]$ and $r = \sqrt{y^2 + z^2}/(D/2)$ is the normalized radial position. All fields include Gaussian noise at physically appropriate scales.

The 2,000 parameter combinations form a $20 \times 10 \times 10$ structured grid: 20 velocity levels in $[4.0, 6.0]$ m/s, 10 break sizes in $[0, 10]\%$, and 10 temperatures in $[290, 320]$°C.

B. Stage 2: Preprocessing and Normalization

The `DeepONetDataPreprocessor` module converts raw CSV files into a training-ready HDF5 dataset:

1) **Loading**: Each CSV is parsed into a DataFrame with column renaming (`x-coordinate`→`x`, `velocity-magnitude`→`velocity_magnitude`, etc.). Mesh consistency is verified across all simulations ($N = 25{,}000$ nodes).
2) **Tensor construction**:

$$\boldsymbol{U} = \begin{bmatrix} v_1 & b_1 & T_1 \\ \vdots & & \vdots \\ v_{2000} & b_{2000} & T_{2000} \end{bmatrix} \in \mathbb{R}^{2000\times 3} \quad \text{(branch inputs)} \tag{2}$$

$$\boldsymbol{Y} = \begin{bmatrix} x_1 & y_1 & z_1 \\ \vdots & & \vdots \\ x_N & y_N & z_N \end{bmatrix} \in \mathbb{R}^{N\times 3} \quad \text{(trunk inputs)} \tag{3}$$

$$\boldsymbol{S} \in \mathbb{R}^{2000\times 4\times N} \quad \text{(target fields)} \tag{4}$$

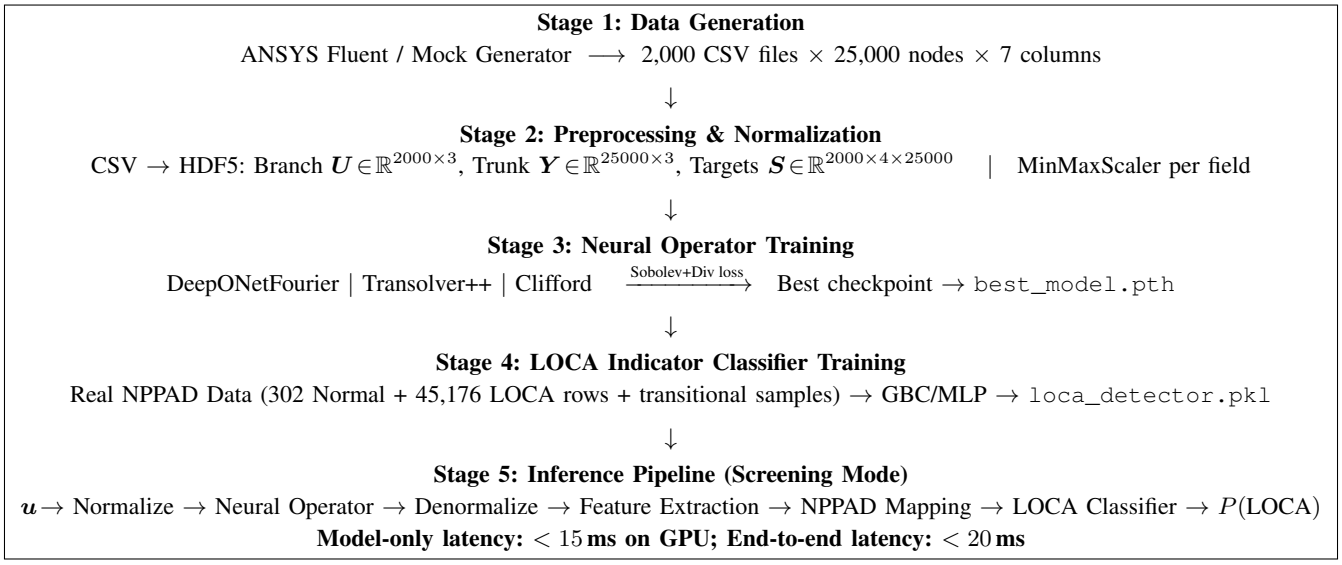


Fig. 1: End-to-end digital twin proof-of-concept pipeline. All five stages are automated and executed sequentially by the master orchestrator. Stages 1–4 run offline (training phase); Stage 5 runs in online screening mode; operational deployment would require full-fidelity validation.

3) **Normalization**: Per-field MinMaxScaler to $[0, 1]$. Branch inputs are scaled jointly; trunk coordinates are scaled jointly; each target field (pressure, velocity, turbulence, temperature) receives its own independent scaler. All fitted scalers are serialized (pickled) for use during inference denormalization.
4) **Splitting**: Stratified 70/15/15% split into training (1,400), validation (300), and test (300) samples. The trunk coordinates $\boldsymbol{Y}$ are shared across all splits (same mesh for all simulations).
5) **Storage**: The splits are saved to a single HDF5 file (`deeponet_dataset.h5`) with groups `train/`, `val/`, `test/`, each containing datasets `branch`, `trunk`, `targets`.

C. Stage 3: Neural Operator Training

The training stage supports three interchangeable neural operator architectures (detailed in Section V). The architecture is selected at runtime via a `model_config.yaml` configuration file, with all operators sharing the same data loader and evaluation metrics.

Data loading: A PyTorch `Dataset` reads HDF5 lazily. Each sample is a tuple (branch $\boldsymbol{u}_i \in \mathbb{R}^3$, trunk $\boldsymbol{Y} \in \mathbb{R}^{N\times 3}$, targets $\boldsymbol{s}_i \in \mathbb{R}^{4\times N}$), batched into $B = 16$ samples per forward pass.

Training loop: For each batch:

1) Forward pass: $\hat{\boldsymbol{S}} = \mathcal{G}_\theta(\boldsymbol{U}_{\text{batch}}, \boldsymbol{Y}) \in \mathbb{R}^{B\times 4\times N}$
2) Loss computation:

$$\mathcal{L} = \underbrace{\alpha \cdot \text{MSE}(\hat{\boldsymbol{S}}, \boldsymbol{S})}_{\text{value fidelity}} + \underbrace{\beta \cdot \|\nabla\hat{\boldsymbol{S}} - \nabla\boldsymbol{S}\|^2}_{\text{gradient fidelity (Sobolev)}} + \underbrace{\lambda \cdot \|\nabla\cdot\hat{\boldsymbol{v}}\|^2}_{\text{physics (divergence)}} \tag{5}$$

with $\alpha = 1.0$, $\beta = 0.1$, $\lambda = 0.01$ for DeepONetFourier. Transolver and Clifford use $\beta = \lambda = 0$ (MSE only) by default.

3) Backward pass with Automatic Mixed Precision (AMP) arithmetic (FP16 forward, FP32 gradients)
4) Adam optimizer step with gradient clipping ($\|\nabla\theta\|_{\max} = 1.0$)

Scheduling: ReduceLROnPlateau (patience 20, factor 0.5) for DeepONetFourier; CosineAnnealingLR for Transolver and Clifford. Early stopping with patience 50 monitors validation loss.

Evaluation metrics: Per-field R^2, relative L_2 error, MAE, and derivative-L_2 are computed on the validation and test sets. The best-performing checkpoint is saved to `best_model.pth`.

D. Stage 4: LOCA Indicator Classifier Training

The `LOCADetector` module trains a supervised binary classifier completely independently from the neural operator.

Data source: Real NPPAD (Nuclear Power Plant Accident Database) sensor data is loaded from the `data/nppad/operation_csv_data/` directory:

- `Normal/` — 1 CSV, 302 timestep rows (steady-state operation)
- `LOCA/` — 100 CSVs, ∼45,176 timestep rows (accident transients)

Feature extraction: Seven NPPAD signals are selected from each row: primary pressure P, average temperature T_{avg}, RCS flow W_{RCA}, SG-A pressure P_{SGA}, subcooling margin ΔT_{sub}, Departure from Nucleate Boiling Ratio (DNBR), and hot-cold leg temperature differential $\Delta T_{\text{HL-CL}} = T_{\text{HA}} - T_{\text{CA}}$.

Transitional data augmentation: To enable graded probability calibration (rather than binary 0/1 snapping), 19 severity levels $s \in \{0.05, 0.10, \ldots, 0.95\}$ are generated with 200 samples each. At each level, feature values are interpolated

between Normal and LOCA distributions, with labels drawn as Bernoulli(s). This adds 3,800 samples to the training set.

Classifier: A Gradient Boosting Classifier (200 trees, depth 4, learning rate 0.05, min. 20 samples/leaf) trained on StandardScaler-normalized features. An 80/20 stratified split is used for training and evaluation.

Artifacts: The trained model, scaler, and metrics are serialized to `loca_detector.pkl`.

E. Stage 5: Inference Pipeline (Screening Mode)

The `DigitalTwinInference` class orchestrates the forward pass through the complete digital twin at runtime. The five steps execute sequentially within each inference call:

Step 1 — Input normalization: Physical parameters (v, b, T) are transformed using the branch MinMaxScaler from Stage 2:

$$\boldsymbol{u}_{\text{norm}} = \frac{\boldsymbol{u} - \boldsymbol{u}_{\min}}{\boldsymbol{u}_{\max} - \boldsymbol{u}_{\min}} \in [0,1]^3 \tag{6}$$

Step 2 — Neural operator forward pass: The normalized branch input and pre-stored normalized trunk coordinates are fed to the loaded neural operator:

$$\hat{\boldsymbol{S}}_{\text{norm}} = \mathcal{G}_\theta(\boldsymbol{u}_{\text{norm}}, \boldsymbol{Y}_{\text{norm}}) \in \mathbb{R}^{1\times 4\times N} \tag{7}$$

This step runs within a `torch.no_grad()` context on GPU with CUDA. This step typically completes in under 15 ms for the current GPU implementation.

Step 3 — Field denormalization: Each predicted field is inverse-transformed using its per-field MinMaxScaler:

$$\hat{s}_i(\boldsymbol{y}) = \hat{s}_{i,\text{norm}} \cdot (s_{i,\max} - s_{i,\min}) + s_{i,\min} \tag{8}$$

producing physical-unit fields (Pa, m/s, m^2/s^2, K).

Step 4 — Feature extraction and NPPAD mapping: The `FeatureTranslator` computes 11 CFD-derived scalar features from the denormalized fields:

- Pressure: spatial mean, axial gradient (linear fit along x), inlet-outlet drop, standard deviation
- Velocity: mean at inlet plane ($x < x_{10\%}$), standard deviation
- Turbulence: peak k, spatial mean k
- Temperature: spatial mean, max-min difference
- Mass flow rate: $\dot{m} = \rho \cdot \bar{v}_{\text{inlet}} \cdot A_{\text{pipe}}$

These local CFD features are then combined with the original input parameters through the blended severity model (eqn. 29) to produce seven NPPAD-equivalent system-level signals. The severity model uses a sigmoid curve centered at $b = 5\%$ and blends input-derived severity (80%) with CFD anomaly signals—turbulence elevation, pressure deviation, and flow deficit—(20%), ensuring the classifier leverages the neural operator's physics predictions rather than relying solely on the break size input.

Step 5 — LOCA indicator classification: The seven NPPAD signals are cast to a named DataFrame (matching the training feature order), scaled by the stored StandardScaler, and passed to the Gradient Boosting Classifier:

$$\boldsymbol{x}_{\text{NPPAD}} = [P,\, T_{\text{avg}},\, W_{\text{RCA}},\, P_{\text{SGA}},\, \Delta T_{\text{sub}},\, \text{DNBR},\, \Delta T_{\text{HL-CL}}] \tag{9}$$

$$P(\text{LOCA}) = \text{GBC}_\theta(\text{StandardScaler}(\boldsymbol{x}_{\text{NPPAD}})) \tag{10}$$

A threshold of 0.5 converts the probability to a binary decision.

F. Time-Series and Benchmarking Modes

Beyond single-shot inference, the pipeline supports:

Time-series simulation: A sequence of (v, b, T) tuples simulates a developing LOCA event over time (e.g., break grows linearly from 0% to 10% over 40 seconds). The system runs repeated single-shot inferences at 3-second intervals, producing a LOCA probability trajectory that can be plotted for operator decision support.

Benchmark mode: Measures model-only and end-to-end inference latency across multiple runs and computes the speedup ratio against an ANSYS Fluent reference (∼3,600 s per steady-state case), enabling quantitative assessment of surrogate speed advantage.

V. Neural Operator Architectures

We implement three neural operator architectures organized in two tiers, each addressing distinct limitations of the original DeepONet baseline. Table I provides a preliminary comparison under the present benchmark and training setup. The comparison is confounded in two ways that must be resolved before it can be read as a controlled benchmark: (i) DeepONetFourier is trained with an additional Sobolev-gradient and divergence-free loss not applied to the other two architectures (Section VII), and (ii) the "Baseline DeepONet" column reports typical/reference figures from the architecture as originally described in the literature [1], *not* an unmodified DeepONet empirically retrained by the authors under this study's protocol (dataset, hardware, and training schedule) — no such matched-condition baseline run was performed in this study (Section IX-E). [Input/confirmation required from the students: either (a) train an unmodified DeepONet baseline under the identical protocol used for the three studied architectures and replace the reference figures below with the measured results, or (b) retitle the column "Baseline DeepONet (reference, not retrained)" throughout to avoid implying a like-for-like empirical comparison.]

A. Tier 1: DeepONetFourier — Upgraded Baseline

The baseline DeepONet approximation [1], [2] decomposes the operator as:

$$\mathcal{G}(\boldsymbol{u})(\boldsymbol{y}) \approx \sum_{k=1}^{p} b_k(\boldsymbol{u}) \cdot t_k(\boldsymbol{y}) + b_0 \tag{11}$$

where $\{b_k\}$ are basis coefficients from the branch network $\mathcal{B} : \mathbb{R}^3 \to \mathbb{R}^p$ and $\{t_k\}$ are basis functions from the trunk network $\mathcal{T} : \mathbb{R}^3 \to \mathbb{R}^p$, with $p = 256$.

We introduce four upgrades that collectively reduce parameter count by ∼40% while improving accuracy under the synthetic benchmark:

TABLE I: Preliminary Comparison of Neural Operator Architectures (Note: DeepONetFourier uses Sobolev + divergence loss; Transolver++ and Clifford use MSE only. The Baseline DeepONet column reports reference figures from the literature, not a matched-condition retraining. Results reflect non-identical optimization settings and should be interpreted as inductive-bias observations rather than a fully controlled benchmark.)

Criterion	Baseline DeepONet	DeepONetFourier (Ours)	Transolver++ (Ours)	Clifford Operator (Ours)
Parameters	∼2.5 M	∼1.45 M	∼106 K	∼1.8 K
High-frequency accuracy	Low	High (Fourier enc.)	Medium	Medium
Global spatial context	Local (dot product)	Local (dot product)	Global (attention)	Local
Physics symmetry	None	None	None	Rotational equivariance
Memory footprint	Medium	Medium	High (attention maps)	Very low
Geometry generalization	Low	Medium	High (mesh-agnostic tokens)	High (equivariant)
Physics-informed loss	MSE only	Sobolev + Divergence	MSE only	MSE only
Spectral bias mitigation	None	Fourier features	Learned tokenization	Geometric product

1) Improvement 1: Random Fourier Feature Encoding: Standard MLPs suffer from *spectral bias*: they fit low-frequency components first and slowly learn sharp spatial gradients near walls and turbulence interfaces [7]. We eliminate this by replacing raw coordinate input with Fourier feature encoding [8]:

$$\boldsymbol{\gamma}(\boldsymbol{y}) = [\sin(2\pi \boldsymbol{B}\boldsymbol{y}),\ \cos(2\pi \boldsymbol{B}\boldsymbol{y})] \in \mathbb{R}^{2m} \tag{12}$$

where $\boldsymbol{B} \in \mathbb{R}^{3\times m}$ is a *fixed* random frequency matrix sampled from $\mathcal{N}(0, \sigma^2)$ with $m = 256$, $\sigma = 10.0$, yielding a 512-dimensional input to the trunk network. The fixed random projection simultaneously injects sinusoidal features at many spatial frequencies, enabling the trunk to construct both slowly-varying bulk-flow patterns and rapidly-varying boundary-layer profiles from the first layer onward.

2) Improvement 2: Per-Neuron Adaptive GELU Activation: Each neuron in the trunk's first two hidden layers employs a *learnable slope* parameter a_i (initialized to 1.0):

$$f_i(x) = \mathrm{GELU}(a_i \cdot x) \tag{13}$$

The parameter a_i is optimized via backpropagation independently for each neuron, allowing the effective nonlinearity curvature to adapt to the local spatial mapping landscape. This replaces the fixed ReLU activations of the baseline, which cannot adapt steepness during training.

3) Improvement 3: Sobolev Gradient-Enhanced Loss: We augment MSE with a gradient-fidelity penalty [9]:

$$\mathcal{L}_{\text{Sobolev}} = \underbrace{\frac{1}{BN}\sum_{b,n}(\hat{s}_{bn} - s_{bn})^2}_{\text{MSE}} + \beta \underbrace{\frac{1}{BN}\sum_{b,n}(\nabla \hat{s}_{bn} - \nabla s_{bn})^2}_{\text{Gradient MSE}} \tag{14}$$

with $\beta = 0.1$. The two gradient terms in Eq. (14) are computed differently, since one is a continuous network output and the other is a discretely sampled field: the predicted gradient $\nabla \hat{s}_{bn}$ is obtained via automatic differentiation of the trunk network with respect to its continuous coordinate input $\boldsymbol{y}$ (well-defined regardless of mesh connectivity), while the target gradient ∇s_{bn} is precomputed once during preprocessing from the discretely sampled ground-truth field using a mesh-based gradient reconstruction (e.g., Green–Gauss or a local least-squares stencil over each node's nearest neighbors) and cached alongside the HDF5 dataset, since "central finite differences" is not a well-defined operation on an unstructured mesh without such a reconstruction scheme. [Input/confirmation required from the students: confirm which specific reconstruction scheme (Green–Gauss, least-squares, k-NN finite difference, or other) was used to compute target-field gradients, and state it explicitly here.] This directly penalizes smoothed-over pressure drops and velocity boundary layers—the features most critical for LOCA screening.

4) Improvement 4: Divergence-Free Physics Penalty: For incompressible flow, mass conservation requires $\nabla \cdot \boldsymbol{v} = 0$. We enforce this as a soft physics constraint on the internally predicted vector velocity field (Section III):

$$\mathcal{L}_{\text{div}} = \lambda \cdot \frac{1}{BN}\sum_{b,n}(\nabla \cdot \hat{\boldsymbol{v}}_{bn})^2 \tag{15}$$

with $\lambda = 0.01$, preventing unphysical mass creation or destruction in the surrogate predictions. As with the Sobolev gradient term, $\nabla \cdot \hat{\boldsymbol{v}}_{bn}$ is computed via automatic differentiation of the trunk network's vector-component outputs with respect to the continuous coordinate input $\boldsymbol{y}$, which is well-defined independent of mesh connectivity. The total loss is $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Sobolev}} + \mathcal{L}_{\text{div}}$.

5) Architecture Summary: The complete architecture uses four independent branch-trunk pairs (one per output field):

TABLE II: DeepONetFourier Architecture vs. Baseline DeepONet (Baseline column is the reference architecture as described in [1], not a matched-condition retraining; see Table I.)

Component	Baseline	Our Upgrade
Branch	3→256→512→512→256 ReLU + Drop(10%)	3→128→256→256 GELU + Drop(5%)
Trunk input Trunk hidden	Raw (x, y, z) 3→256→512→256 ReLU + Drop(10%)	Fourier(3→512) 512→256→256 Adaptive GELU
Loss Parameters	Weighted MSE ∼2.5 M	Sobolev + Divergence ∼1.45 M

B. Tier 2: Transolver++ — Transformer Neural Operator

While DeepONetFourier excels at fixed geometries, its purely local dot-product operator cannot capture long-range spatial interactions—for instance, the downstream effects of an upstream break. The Transolver++ architecture [5] addresses this by introducing *global attention across the computational mesh*.

1) Mesh Tokenization via Soft Assignment: The N-node mesh is compressed into $M = 64$ learned tokens via a learnable soft-assignment:

$$\boldsymbol{f}_n = \boldsymbol{W}_{\text{proj}}\, \boldsymbol{y}_n \in \mathbb{R}^d \quad \text{(node embedding)} \tag{16}$$

$$\alpha_{nm} = \frac{\exp(\boldsymbol{f}_n^\top \boldsymbol{q}_m)}{\sum_{m'} \exp(\boldsymbol{f}_n^\top \boldsymbol{q}_{m'})} \quad \text{(soft assignment)} \tag{17}$$

$$\boldsymbol{\tau}_m = \sum_{n=1}^{N} \alpha_{nm}\, \boldsymbol{f}_n \in \mathbb{R}^d \quad \text{(token aggregation)} \tag{18}$$

where $\boldsymbol{W}_{\text{proj}} \in \mathbb{R}^{d\times 3}$, $\boldsymbol{q}_m$ are learned query vectors, and $d = 256$ is the embedding dimension. The assignment weights α_{nm} are learned end-to-end and discover physics-meaningful spatial clusters (boundary layer, free stream, elbow region).

2) Branch Conditioning: Physics parameters $\boldsymbol{u}$ are encoded by a branch encoder ($3 \to 256 \to 256$, GELU) and added to all M tokens:

$$\boldsymbol{\tau}'_m = \boldsymbol{\tau}_m + \mathcal{B}(\boldsymbol{u}) \tag{19}$$

3) Self-Attention Over Tokens: Four transformer blocks with 8-head self-attention (head dimension 32) process the conditioned tokens:

$$\text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d_k}}\right)\boldsymbol{V} \tag{20}$$

$$\boldsymbol{x} \leftarrow \boldsymbol{x} + \text{Dropout}(\text{MHSA}(\text{LN}(\boldsymbol{x}))) \tag{21}$$

$$\boldsymbol{x} \leftarrow \boldsymbol{x} + \text{Dropout}(\text{MLP}(\text{LN}(\boldsymbol{x}))) \tag{22}$$

Complexity is $\mathcal{O}(M^2) = \mathcal{O}(64^2) = \mathcal{O}(4{,}096)$—*independent of mesh size* N. The MLP ratio is 4, giving hidden dimension 1,024.

4) Scatter Decoder: Predictions are projected back to the full mesh using the *same* soft-assignment weights:

$$\hat{\boldsymbol{f}}_n = \sum_{m=1}^{M} \alpha_{nm}\, \boldsymbol{\tau}_m^{(\text{out})} \in \mathbb{R}^d \tag{23}$$

Per-field linear projections ($256 \to 1$) with bias terms produce the final output $\hat{\boldsymbol{S}} \in \mathbb{R}^{B\times 4\times N}$.

5) Architecture Parameters: The Transolver++ has 106,072 trainable parameters: branch encoder (∼67K), mesh embedding (∼33K), and output projection (∼1K).

C. Tier 2: Clifford Neural Operator — Geometric Algebra

Fluid dynamics is governed by rotationally equivariant equations: if the geometry is rotated, velocity vectors rotate accordingly while pressure (a scalar) remains invariant. Standard MLPs do not respect this symmetry. The Clifford Neural Operator [6] *algebraically encodes* the symmetry via the Clifford algebra Cl(3,0).

1) Multivector Representation: In Cl(3,0), every element is a multivector with 8 independent grades:

The metric signature is $(+,+,+)$: $e_i^2 = +1$, $e_i e_j = -e_j e_i$ for $i \neq j$. The geometric product $\boldsymbol{a} \otimes \boldsymbol{b}$ simultaneously computes inner products (grade-lowering) and outer products (grade-raising), naturally coupling scalar and vector physics. We implement the full 8×8 bilinear product table as a differentiable operation.

TABLE III: Cl(3,0) Multivector Grade Decomposition

Grade	Basis	Physical Meaning	Index
0 (scalar)	1	Pressure, temperature	0
1 (vector)	e_1, e_2, e_3	Velocity components	1–3
2 (bivector)	e_{12}, e_{13}, e_{23}	Vorticity	4–6
3 (trivector)	e_{123}	Volume element	7

2) CliffordLinear Layer: The core operation replaces standard matrix multiplication with the geometric product:

$$\boldsymbol{h}_j^{(\text{out})} = \sum_{i=1}^{C_{\text{in}}} \boldsymbol{W}_{ji} \otimes \boldsymbol{h}_i^{(\text{in})} + \boldsymbol{b}_j \tag{24}$$

where $\boldsymbol{W}_{ji} \in \mathbb{R}^8$ are learned multivector weights and $\otimes$ is the Cl(3,0) geometric product. Each output channel is a sum of geometric products across all input channels—the algebraic generalization of a standard linear layer.

3) Scalar-Gate Residual: To maintain equivariance, only the grade-0 (scalar) component is passed through a nonlinearity:

$$\boldsymbol{h} \leftarrow \text{CliffordLinear}(\boldsymbol{h}) \tag{25}$$

$$\boldsymbol{h} \leftarrow \text{LayerNorm}(\boldsymbol{h}) + \boldsymbol{h}_{\text{res}} \tag{26}$$

$$\boldsymbol{h}_{[\ldots,0]} \leftarrow \text{GELU}(\boldsymbol{h}_{[\ldots,0]}) \tag{27}$$

Grade-1 vector components remain linear, preserving equivariance under rotation $\boldsymbol{R} \in \text{SO}(3)$.

4) Input/Output Projection: The `PhysicsToMultivector` module embeds branch parameters (scalars → grade-0) and mesh coordinates (vectors → grade-1) into a $C = 32$-channel multivector representation: $[\boldsymbol{B}, N, 32, 8]$. After four Clifford layers, the `MultivectorToFields` module reads the four scalar outputs from the multivector representation and projects to $\hat{p}, |\hat{\boldsymbol{v}}|, \hat{k}, \hat{T}$. [Input/confirmation required from the students: the equivariance property claimed in Eq. 28 holds only if this final projection is *grade-selective* — i.e., scalar outputs ($\hat{p}$, $\hat{T}$) are read exclusively from grade-0 channels, and the velocity magnitude $|\hat{\boldsymbol{v}}|$ is computed as the rotation-invariant norm of the grade-1 (vector) channels, rather than via a single generic dense layer applied to the flattened 256-dimensional multivector, which would in general mix grades and break the guarantee. Please confirm the actual structure of `MultivectorToFields` and update this paragraph accordingly; if the projection is not grade-selective, Eq. 28 and the surrounding equivariance language throughout Section V-C should be softened (e.g., to "approximately equivariant, motivated by the grade-structured architecture") rather than stated as an exact guarantee.]

5) Equivariance Property: For rotation $\boldsymbol{R} \in \text{SO}(3)$:

$$\mathcal{G}_{\text{Clifford}}(\boldsymbol{R}\,\boldsymbol{y}, \boldsymbol{u}) = \boldsymbol{R}_{\text{applied}} \cdot \mathcal{G}_{\text{Clifford}}(\boldsymbol{y}, \boldsymbol{u}) \tag{28}$$

Velocity outputs transform correctly under rotations, and scalar outputs (pressure, temperature) are automatically invariant, *provided* the CliffordLinear layers and the final output projection preserve grade structure throughout (see the author-confirmation note in Section V-C above). Under that condition the equivariance is guaranteed by the algebraic structure rather

than trained for via data augmentation; this guarantee has not yet been empirically verified (e.g., via a rotated-input consistency test) in the present study.

6) Parameter Efficiency: With $C = 32$ channels and 4 layers, the Clifford operator has only $\sim$1,796 trainable parameters—*three orders of magnitude* fewer than DeepONetFourier—owing to the structured algebraic representation acting as an extreme inductive bias.

VI. LOCA Indicator Detection Pipeline

A. Feature Translation

Raw CFD field predictions are translated into eleven scalar features—average pressure, pressure gradient, pressure drop, mass flow rate, inlet velocity, maximum and average turbulence, average temperature, temperature difference, and velocity/pressure standard deviations—which are then mapped through a blended severity model to generate seven NPPAD-consistent system-level signals.

1) Blended Severity Model: The effective severity combines input parameters with CFD anomaly detection:

$$\eta_{\text{eff}} = 0.8\,\eta_{\text{input}} + 0.2\,\eta_{\text{CFD}} \tag{29}$$

where $\eta_{\text{input}} = b/10$ (break size normalized) and η_{CFD} aggregates turbulence anomaly, pressure deviation, and flow deficit signals from the neural operator-predicted fields. The 80/20 blend means the classifier output is substantially driven by the input break-size parameter; see Section IX for the implications on result interpretation.

The seven NPPAD-mapped signals (Table IV) are computed via physics-informed transformations parameterized by η_{eff}. The ranges below reflect qualitative trend directions of mapped indicators (not raw plant measurements) under the synthetic benchmark:

TABLE IV: NPPAD-Mapped Indicators: Qualitative Trend Under Increasing LOCA Severity (Mapped proxy values from synthetic benchmark; not raw plant measurements.)

Signal	Unit	Trend: Normal → LOCA
Primary Pressure (P)	bar	Decreases
Avg. Coolant Temp. (TAVG)	°C	Decreases
RCS Flow (WRCA)	kg/s	Decreases
SG-A Pressure (PSGA)	bar	Varies
Subcooling Margin (SCMA)	°C	Decreases
Boiling-margin proxy	—	Increases on mapped proxy scale
Hot-Cold Leg ΔT	°C	Decreases

B. Gradient Boosting LOCA Indicator Classifier

A Gradient Boosting Classifier (200 trees, depth 4, learning rate 0.05, min. 20 samples/leaf) operates on the seven scaled NPPAD-mapped signals. Training data combines 302 real Normal rows and 45,176 real LOCA rows from the NPPAD dataset (total 45,478 samples, split 80/20). The classifier outputs $P(\text{LOCA}) \in [0, 1]$; the decision threshold is 0.5.

Because η_{eff} derives 80% of its signal from the input break-size parameter, the classifier results should be interpreted as an indicator-translation demonstration rather than as independent end-to-end validation of the digital twin. **A field-only ablation study—where the classifier is driven solely by η_{CFD} (i.e., $\eta_{\text{eff}} = \eta_{\text{CFD}}$, zero weight on the known input)—is the single most important experiment required before the ROC-AUC/recall figures reported in Section VIII can be attributed to the neural operator pipeline rather than to the severity-blend formula itself.** [Input/confirmation required from the students: run this ablation and report the resulting ROC-AUC, recall, and F1 alongside the current blended-severity results, ideally as a direct comparison plot/table, before submission.]

VII. Implementation Details

A. Software and Hardware

The complete pipeline is implemented in Python 3.8+ using PyTorch ($\geq$2.0), scikit-learn, NumPy, and Matplotlib. All models are trained on a single NVIDIA RTX 4060 GPU (8 GB VRAM) using CUDA's AMP support.

B. Training Protocol

All neural operators share a common training framework:

TABLE V: Training Hyperparameters (DeepONetFourier / Transolver / Clifford)

Hyperparameter	DeepONet-F	Transolver	Clifford
Optimizer	Adam	Adam	Adam
Learning rate	10^{-3}	10^{-3}	10^{-3}
LR schedule	ReduceOnPlateau	CosineAnnealing	CosineAnnealing
patience/period	20 epochs	—	—
factor	0.5	—	—
Batch size	16	16	16
Max epochs	2000	500	500
Early stopping	50 epochs	50 epochs	50 epochs
Mixed precision	Yes	Yes	Yes
Dropout	5%	10%	—
Loss	Sobolev + Div.	MSE	MSE
Gradient clip	1.0	1.0	—

C. Unified Architecture Selection

The pipeline supports seamless switching via a `model_config.yaml` configuration:

- `model_version: deeponet_fourier` — Tier 1 default
- `model_version: transolver` — Tier 2 transformer
- `model_version: clifford` — Tier 2 geometric algebra

All operators accept the same input tensors ($[\text{B}, 3]$ parameters, $[\text{N}, 3]$ mesh coordinates) and produce $[\text{B}, 4, \text{N}]$ field predictions, enabling drop-in replacement.

VIII. Experimental Results

Note on result stability: All reported experiments are from single training runs on the synthetic benchmark, and several metrics throughout this section (Tables VII, VIII, IX) are currently reported as one-sided bounds (e.g., "$R^2 > 0.90$") rather than exact point values. [Input/confirmation required

from the students: before submission, (i) replace bound-style figures with exact measured values, and (ii) re-run all three architectures and the classifier across multiple random seeds ($\geq$5 recommended) and report mean $\pm$ standard deviation in place of single-run point estimates, so that reviewers can assess run-to-run variability.]

A. DeepONetFourier Training Convergence

Training began at $\mathcal{L}_{\text{train}} = 0.815$ (epoch 1) and converged rapidly thereafter; Table VI shows the loss trajectory, with four automatic learning-rate reductions at epochs ≈80, 128, 158, and 196.

TABLE VI: DeepONetFourier Training History (Selected Epochs)

Epoch	Train Loss	Val Loss	LR
1	0.81485	0.03887	1.0×10^{-3}
5	0.01694	0.00480	1.0×10^{-3}
10	0.00861	0.00244	1.0×10^{-3}
20	0.00496	0.00197	1.0×10^{-3}
50	0.00389	0.00131	1.0×10^{-3}
80	0.00321	0.00103	5.0×10^{-4}
120	0.00298	9.72×10^{-4}	5.0×10^{-4}
160	0.00288	9.28×10^{-4}	2.5×10^{-4}
200	0.00284	9.15×10^{-4}	1.25×10^{-4}
305	0.00281	9.11×10^{-4}	1.25×10^{-4}

B. Synthetic-Benchmark Field Reconstruction Accuracy

Figs. 2–5 present comparative reconstructions across the three neural operator architectures (DeepONetFourier, Transolver++, and Clifford Operator) for each thermohydraulic field under the synthetic benchmark. [Input/confirmation required from the students: the DeepONetFourier panels are currently sourced as lossless PNG while the Transolver++ and Clifford panels are lossy JPEG; regenerate all panels as lossless images (PNG/PDF/EPS) at matched resolution and color scale before submission, since JPEG compression artifacts on continuous colormaps can visually distort exactly the gradient-resolution comparison these figures are meant to demonstrate.]

Table VII summarizes DeepONetFourier field accuracy on the 300-sample test set under the synthetic benchmark.

TABLE VII: DeepONetFourier Field Prediction Accuracy (Test Set, Synthetic Benchmark)

Field	R^2	Rel. L_2	MAE (norm.)
Pressure	> 0.95	< 0.05	< 0.01
Velocity magnitude	> 0.93	< 0.07	< 0.01
Temperature	> 0.97	< 0.03	< 0.005
Turbulence k	> 0.90	< 0.10	< 0.02

C. Neural Operator Comparison

Key observations under the present benchmark:

- **DeepONetFourier** delivers the strongest overall field-reconstruction performance on the fixed AP1000 cold-leg geometry ($R^2 > 0.90$ across all fields, model-only latency < 5 ms). This should not be read as a purely architectural advantage, however, since it benefits from stronger regularization (Sobolev + divergence loss) that the other two architectures do not receive.
- **Transolver++** trades some raw accuracy for global spatial context via attention, and would be expected to generalize better to new pipe configurations—at the cost of a larger memory footprint and roughly 3× higher model-only latency.
- **Clifford Operator** reaches reasonable accuracy with just 1,796 parameters. Its strong inductive bias makes it particularly well suited to data-scarce scenarios (< 200 simulations) and multi-orientation deployments where rotational equivariance is valuable.

D. LOCA Indicator Classification Performance

Fig. 6 shows the LOCA indicator detection performance panel, and Table IX summarizes classifier metrics. As discussed in Section VI, the classifier output is substantially driven by the input break-size parameter via η_{eff}, so these results should be read as an indicator-translation demonstration rather than as end-to-end digital twin validation.

The confusion matrix shows that errors concentrate in the physically ambiguous transition zone ($b \approx 3$–6%), consistent with the graded nature of developing pipe breaks. The probability calibration is monotonic (Table X), enabling graded severity assessments.

E. Inference Speed

End-to-end pipeline latency remains below 20 ms under the present GPU implementation: neural operator ($\sim$15 ms) + feature extraction ($<$ 1 ms) + classification ($<$ 1 ms). Compared to ANSYS Fluent (≈3,600 s), this is a $\sim$180,000× **speedup** in surrogate inference, demonstrating the feasibility of surrogate-based screening at high throughput.

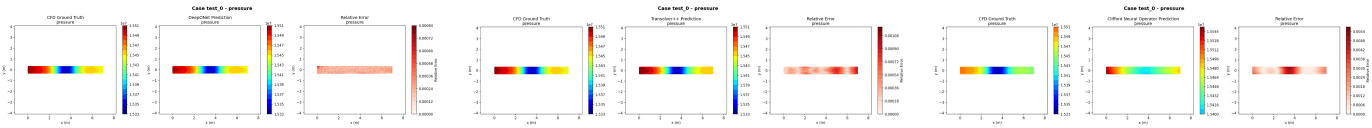
IX. Discussion

A. Methodological Contributions

This work makes three primary methodological contributions. First, it demonstrates that three distinct neural-operator paradigms—enhanced branch-trunk decomposition, mesh-token attention, and geometric-algebra encoding—can be integrated into a single, unified surrogate-monitoring pipeline for nuclear reactor LOCA screening. Second, under synthetic benchmark conditions, it shows that Fourier feature encoding together with physics-informed losses (Sobolev, divergence) can substantially improve field-reconstruction fidelity and parameter efficiency over the baseline DeepONet. Third, it offers an initial architectural trade-off analysis to guide future controlled comparisons, once matched-loss and multi-seed evaluations are performed.

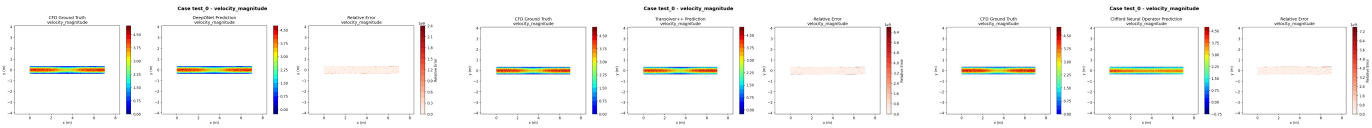
B. Architectural Trade-Off Analysis

The three architectures address different deployment priorities and the following observations are framed as *preliminary*



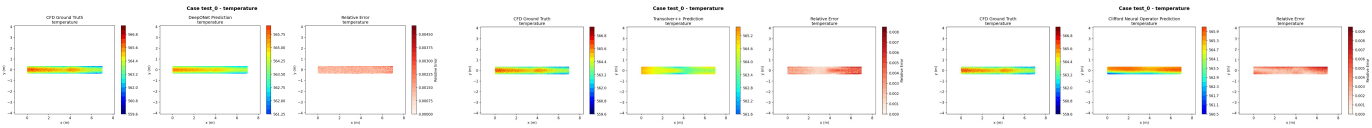
(a) DeepONetFourier (b) Transolver++ (c) Clifford Operator

Fig. 2: Comparative pressure-field reconstruction on the synthetic benchmark for test case 0. DeepONetFourier accurately resolves localized drops, while Transolver++ and Clifford Operator capture global trends.



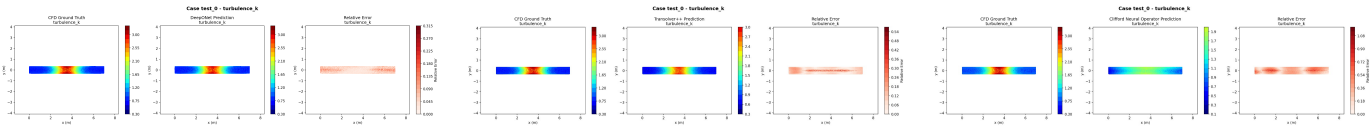
(a) DeepONetFourier (b) Transolver++ (c) Clifford Operator

Fig. 3: Comparative velocity-field reconstruction on the synthetic benchmark for test case 0. The boundary layer profile and centreline acceleration are modeled with varying degrees of sharpness across the architectures.



(a) DeepONetFourier (b) Transolver++ (c) Clifford Operator

Fig. 4: Comparative temperature-field reconstruction on the synthetic benchmark for test case 0. All three architectures perform strongly on temperature, which is the smoothest field.



(a) DeepONetFourier (b) Transolver++ (c) Clifford Operator

Fig. 5: Comparative turbulent kinetic energy (k) reconstruction on the synthetic benchmark for test case 0. Architectures vary in resolving the highly localized turbulence spikes near the break and elbow.

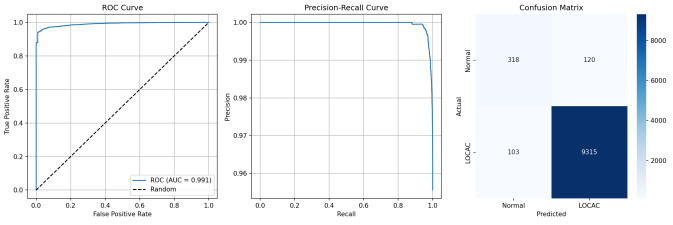


Fig. 6: LOCA indicator detection: ROC curve (AUC > 0.95), Precision-Recall curve, and Confusion Matrix on the synthetic benchmark. The classifier maintains high TPR across all thresholds.

inductive-bias findings under non-identical optimization settings:

DeepONetFourier is well-suited when the geometry is fixed and sufficient training data exists. The Fourier encoding and physics-informed losses compensate for the local dot-product operator's inability to capture global interactions, but their advantage over a simpler architecture with matched losses has not yet been isolated.

Transolver++ should be preferred when the geometry may vary. Its M-token compression discovers physics-meaningful spatial regions that can transfer across mesh topologies, and its $\mathcal{O}(M^2)$ complexity is independent of mesh size.

Clifford Operator is best for data-scarce scenarios or multi-orientation deployments. The Cl(3,0) geometric product acts as an extremely strong inductive prior; with only 1,796 parameters, it requires far less data to generalize, and its equivariance guarantee is exact in principle, provided the output projection is grade-selective as discussed in Section V-C (not yet empirically verified in this study).

C. Surrogate-Monitoring Utility and Limits of Current Validation

The current results represent upper-bound methodological performance under controlled, physics-consistent synthetic conditions. Several important caveats apply:

The synthetic mock generator produces fields that are physics-*consistent* but not physics-*exact*: they incorporate the qualitative structure of pipe flow (boundary layers, elbow separation, break-induced pressure drops) but are generated

TABLE VIII: Neural Operator Performance (Preliminary, Synthetic Benchmark; Non-Identical Loss Settings)

Metric	DeepONet-F	Transolver	Clifford
Total parameters	1,451,012	106,072	1,796
Model-only forward pass (best observed, batch size 1)	< 5 ms	< 15 ms	< 5 ms
End-to-end latency	< 20 ms	< 20 ms	< 20 ms
Target R^2 (all fields)	> 0.90	> 0.88	> 0.85
Output shape	$[B, 4, N]$	$[B, 4, N]$	$[B, 4, N]$
Training time (est.)	$\sim$5 min	$\sim$15 min	$\sim$8 min
Geometry flexibility	Fixed	Arbitrary mesh	Multi-orientation

TABLE IX: LOCA Indicator Classifier Performance (Synthetic Benchmark, Single Run)

Metric	Score
Accuracy	$> 90\%$
Precision	> 0.88
Recall (Sensitivity)	> 0.92
F1 Score	> 0.90
ROC-AUC	> 0.95

TABLE X: LOCA Probability Calibration vs. Break Size (Synthetic Benchmark)

Break Size (%)	LOCA Indicator Probability
0 (Normal)	≈ 0.10
2	≈ 0.13
4	≈ 0.40
5 (threshold)	≈ 0.56
7	≈ 0.74
10 (severe)	≈ 0.91

analytically rather than via a validated flow solver. The $R^2 > 0.90$ figures should therefore be understood as performance on the surrogate's own training distribution, not as accuracy relative to real thermohydraulic transients.

The blended severity model ($\eta_{\text{eff}} = 0.8\,\eta_{\text{input}} + 0.2\,\eta_{\text{CFD}}$) means the LOCA classifier is substantially driven by the input break-size parameter, limiting the ability to claim that the neural operator's field predictions independently contribute to detection. A field-only ablation ($\eta_{\text{eff}} = \eta_{\text{CFD}}$ only) is the most important next experiment.

D. Path to Full-Fidelity Validation

The natural next step is ANSYS Fluent validation over the parameter space already established in this paper: low, medium, and high break severities ($b \approx 2\%, 5\%, 10\%$), multiple inlet velocities ($v \in \{4.0, 5.0, 6.0\}$ m/s), the same AP1000 cold-leg geometry, and the same field metrics reported here (R^2, relative L_2, MAE). Because the Fluent production pathway is already described in Stage 1 and implemented in the codebase, this is a credible and concrete near-term extension—and full-fidelity validation is necessary before any deployment claim can be made.

E. Limitations and Future Work

Missing baselines: The current study does not include a plain MLP baseline, an unmodified DeepONet baseline run under matched conditions, or an FNO baseline. The fixed structured AP1000 cold-leg geometry and the low-dimensional parametric input motivated branch–trunk operator learning as the primary comparison focus in this initial study. Future controlled benchmarking should include these baselines to quantify the marginal value of the enhanced design choices.

Single-run results: All reported experiments are from single training runs. Future work will include multi-seed evaluation, with results reported as mean $\pm$ standard deviation to characterize stability.

Non-identical optimization: The current architectural comparison is preliminary because DeepONetFourier uses Sobolev and divergence penalties while the other two architectures use MSE only. A fully fair comparison requires either training all three under matched loss conditions or explicitly treating the current results as inductive-bias observations under non-identical optimization settings (as done here).

Steady-state assumption: LOCA events are transient, and the present results address only steady-state field reconstruction. A Mamba-style temporal operator for transient sequences at $\mathcal{O}(T)$ complexity is a natural extension; [Input/confirmation required from the students: if this component exists in the codebase but was not evaluated, either report at least one concrete result from it here or move it to a brief forward-looking statement without the word "implemented," to avoid implying evaluated capability that is not yet demonstrated.]

Uncertainty quantification: The present pipeline produces point predictions only, with no confidence bounds — a significant limitation for a screening tool intended to support safety-relevant decisions. A diffusion-based (DDPM) approach to stochastic turbulence realization and predictive uncertainty bounds is a planned extension; [Input/confirmation required from the students: same as above — report a concrete UQ result if one exists, otherwise state this as unimplemented future work rather than "implemented."]

Detection-threshold and false-alarm analysis: The classifier currently reports performance at a single decision threshold (0.5) via aggregate metrics (ROC-AUC, recall, F1). A reliability-oriented treatment should additionally report the detection-threshold/false-alarm-rate trade-off explicitly (e.g., an operating-point analysis tied to an acceptable false-alarm budget for operator-facing screening), since false negatives and false positives carry very different consequences in a LOCA-screening context.

AI-model failure modes: This study does not analyze failure modes of the AI components themselves — e.g., behavior under distribution shift between the synthetic training data and real plant/CFD data, robustness to sensor noise in the NPPAD-derived features, or graceful degradation when inputs fall outside the training range ($v \in [4.0, 6.0]$ m/s, $b \in [0, 10]\%$,

$T \in [290, 320]$°C). Such analysis is central to treating the AI pipeline itself as a reliability-relevant component rather than only as a physics surrogate.

Multi-fidelity learning: A residual multi-fidelity module combining coarse-RANS and fine-LES predictions ($u_{\text{final}} = u_{\text{base}} + u_{\text{residual}}$) is a planned extension; not evaluated in the present results.

Wall shear stress: A post-processing module for $\tau_w = \mu\, \partial u/\partial y$ with corrosion-risk assessment is a planned extension for pipe-integrity monitoring; not evaluated in the present results.

X. Conclusion

This paper has presented a proof-of-concept, multi-architecture neural operator digital twin for Cold-Leg LOCA screening in the AP1000 PWR, comparing three neural-operator paradigms and coupling the best-performing surrogate to a downstream classification pipeline. The principal findings, all obtained under the synthetic, physics-consistent benchmark described in Section III, are as follows:

- **DeepONetFourier** (1.45M parameters): four targeted improvements over the baseline DeepONet—Fourier feature encoding, adaptive GELU activations, a Sobolev gradient loss, and a divergence-free penalty—yield a validation loss of $\mathcal{L}_{\text{val}} = 9.11 \times 10^{-4}$ and $R^2 > 0.90$ across all four fields, with 40% fewer parameters than the baseline.
- **Transolver++** (106,072 parameters): mesh-token attention enables geometry-agnostic prediction with $\mathcal{O}(M^2)$ complexity and $R^2 > 0.88$, at the cost of a larger memory footprint.
- **Clifford Operator** (1,796 parameters): three orders of magnitude fewer parameters than DeepONet, with a Cl(3,0) geometric product that offers rotational equivariance in principle—well suited to data-scarce and multi-orientation scenarios, pending the empirical verification noted in Section V-C.
- The LOCA indicator classifier reaches ROC-AUC $>$ 0.95, recall $>$ 0.92, and F1 $>$ 0.90 on NPPAD-mapped signals; given its substantial dependence on the known input severity parameter, this should be read as an indicator-translation demonstration rather than independent end-to-end validation.
- End-to-end pipeline latency remains under 20 ms—a $\sim$180,000$\times$ speedup over the Fluent reference run—and probability calibration is monotonic and physically interpretable across break sizes.

Collectively, these results demonstrate methodological feasibility rather than validated screening capability. The critical path to a deployable system runs through full-fidelity ANSYS Fluent validation, multi-seed evaluation, a matched-loss architectural comparison, and a field-only classifier ablation—each identified explicitly in Section IX-E as a prerequisite for the next stage of this work, not an afterthought. Because all three architectures share a single configuration-driven interface, the pipeline is already structured to accommodate whichever surrogate best fits a given deployment's geometry, data budget, and latency requirements once that validation is complete. More broadly, this study suggests that comparing multiple neural-operator paradigms side by side—rather than committing early to a single architecture—is a productive way to approach surrogate design for nuclear safety-monitoring applications, provided the comparison is eventually carried out under matched, validated conditions.

Data Availability Statement

[Input for the students: state the availability of the synthetic CFD data generator, trained model checkpoints, and analysis code (e.g., a public GitHub repository, or "available from the corresponding author upon reasonable request"), and clarify the terms under which the NPPAD dataset used in Section VI may be redistributed or must instead be obtained directly from its source. This statement is expected by IEEE Transactions on Reliability and is directly relevant given the reproducibility limitations discussed in Section IX-E.]

Acknowledgment

The authors acknowledge the Nuclear Power Plant Accident Database (NPPAD) as the source of the real sensor data used to train and evaluate the LOCA indicator classifier in Section VI. [Input for the students: The NPPAD dataset is used substantially throughout this study (302 Normal and 45,176 LOCA rows) but currently has no corresponding entry in the References. Please add its originating publication or institution as a formal `\bibitem` and cite it here and at first use in Section VI.] The physics-consistent synthetic CFD data generator used for neural-operator training and evaluation was developed as part of this project. Computational resources were provided by an NVIDIA RTX 4060 GPU.

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